16 The nth term of a sequence is given by $an^2 + bn$ where a and b are integers.

The 2nd term of the sequence is –2
The 4th term of the sequence is 12

(a) Find the 6th term of the sequence.

$n = 2$: $4a + 2b = -2$ (1)

$n = 4$: $16a + 4b = 12$ (2)

1 mark
Substitute a known term

Multiply equation (1) by 2, then subtract from (2).

$$\begin{array}{rl} & 16a + 4b = 12 \\ - & 8a + 4b = -4 \\ \hline \end{array}$$

1 mark
Make b coefficients identical and subtract to eliminate b

Subtract: $8a = 16$

$\div 8$ $\div 8$

$a = 2$; $4(2) + 2b = -2$, so $2b = -10$ and $b = -5$.

1 mark
Both a = 2 and b = −5

Sixth term $= 2 \times 6^2 - 5 \times 6 = 72 - 30 = 42$.

1 mark
Substitute n = 6

42

(4)

Here are the first five terms of a different quadratic sequence.

0 2 6 12 20

(b) Find an expression, in terms of n, for the nth term of this sequence.

First differences: 2 4 6 8

Second differences: 2 2 2

Second difference ÷ 2 = 1, so start with n^2.

1 mark
Identify the quadratic term

Subtract n^2: −1, −2, −3, −4, −5. This remainder is $-n$.

So the nth term is $n^2 - n$.

Check: $n = 1$ gives 0; $n = 5$ gives 20.

$n^2 - n$

(2)

1 mark
Correct complete expression

(Total for Question 16 is 6 marks)